QME Student Handbook

2018–2019

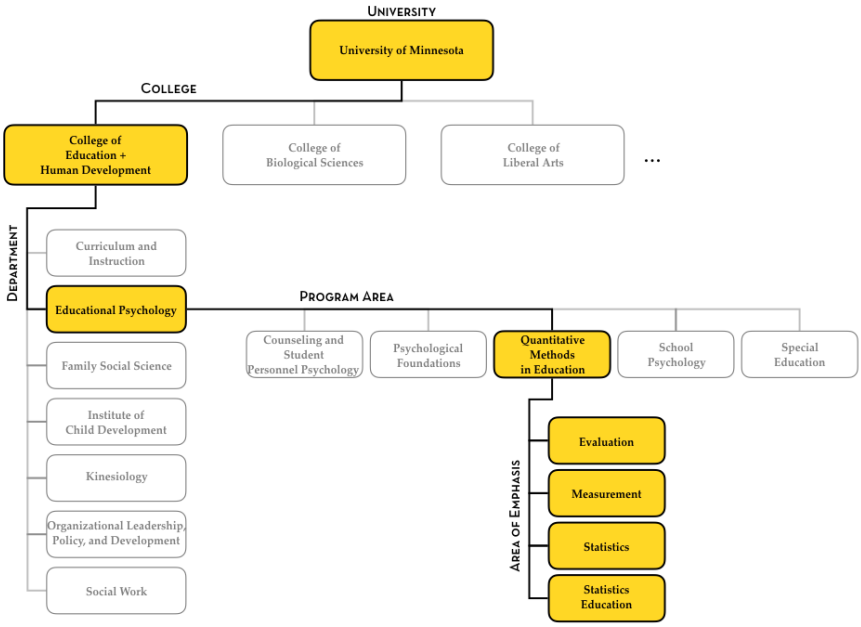
Acknowledgments

The *Educational Psychology Graduate Student Handbook* and the University of Minnesota *Graduate School Catalog* were two resources that were instrumental in updating this handbook.

Thank you to Lori Boucher for explaining the myriad forms, policies, and processes, as well as for taking time to answer endless questions. Also thanks to Scott McConnell for sharing and allowing the use and adaptation of School Psychology's *Faculty Guidelines for Preliminary Oral Examination Papers*. Much of the revised information regarding the oral preliminary examination comes from that document.

Introduction

This handbook is for graduate students admitted to the Quantitative Methods in Education (QME) program. QME is one of five program areas in the Department of Educational Psychology. The Department of Educational Psychology is one of seven departments (or schools) within the College of Education and Human Development at the University of Minnesota.



Graduate students in QME are accountable to the policies described in the *QME Student Handbook* for the year of their admission along with those of the Department of Educational Psychology, Graduate School, College of Education and Human Development and University of Minnesota.

Note: QME students graduate with a degree in Educational Psychology. The program area (QME) will be listed on the transcript, but only Educational Psychology is listed on the actual diploma.

The QME program

QME students learn to use measurement, evaluation, and statistical techniques and models to study educational phenomena. QME students work with faculty to develop new methods for analyzing education data, create assessment tools and evaluation models and methods, and generate innovative approaches to teaching and learning statistics. Graduates of the QME program go on to careers in academia, research and evaluation centers, test publishing firms, public school systems, state departments of education, and private industry.

Areas of Emphasis

QME offers four areas of emphasis: measurement, evaluation, statistics, and statistics education. Students who specialize in measurement study the theory and methods of measuring important variables in educational research, such as achievement, attitude, and specialized cognitive and non-cognitive variables. They learn to develop tests and surveys for specific educational purposes, such as assessing the reading and mathematics achievement of elementary school students, and study issues related to measurement and accountability. Students specializing in evaluation learn to evaluate the effectiveness of educational programs and education-related human services using a variety of quantitative and qualitative techniques, and may also study issues related to measurement and accountability. Students specializing in statistics develop an understanding of the relationship between research design and statistical analysis, acquire skills in using a variety of statistical techniques appropriate for specific problems in education, and may study issues related to teaching statistics. Skills in measurement, evaluation, and statistics prepare graduates for a variety of positions, including test publishing firms, colleges, universities, research and evaluation centers, public school systems, state departments of instruction, and industry. Students specializing in statistics education investigate issues related to teaching and learning statistics and gain experience in statistics instruction.

Regardless of emphasis, the coursework and research projects that all QME students complete give them a broad, but rigorous foundation in both qualitative and quantitative methodologies, as well as in general educational issues. This training gives them the foundational knowledge needed to conduct research on methodologies, help to train others in methodology, or utilize their skills to conduct research in related fields.

Research orientation of the QME program

The QME faculty strongly believes that collaborative and independent research activities should be central to a student's training and that these activities promote the acquisition of competencies essential to success upon attaining a degree. A premium is placed on involving students in research activities, including critically analyzing a body of literature, generating research questions which address specific issues in quantitative methods in education, such as developing empirically-based tests and surveys, designing and executing accountability studies in ways that take into account key methodological considerations (e.g., instrument construction and evaluation), correctly analyzing and interpreting research results, and succinctly communicating in writing the results of research studies. These competencies apply to both the master's and doctoral programs, but doctoral students are further expected to develop the ability to work independently and demonstrate a greater depth of mastery of these competencies. Settings in which students frequently engage in research include collaboration with other students on course-related research, supervised research done under the direction of faculty researchers, and independently conducting a research study.

Academic advising and supervision

At the graduate level, the academic advisor plays a much greater role than they typically do at the undergraduate level. In addition to helping students navigate the coursework and milestones in the student's degree program, they often play a pivotal role in helping students form a basis of content and research knowledge.

When a student is accepted for admission in the QME program, a QME faculty member is assigned as the student's academic advisor. The advisor acts as a liaison between the student and the faculty and also helps the student. In the QME program, academic advisors:

1. Supervise the development of the student's degree program;
2. Approve all coursework;
3. Annually review the student's academic progress and notify the student if progress is unsatisfactory; and
4. Certify that graduation requirements have been met.

Students should recognize that there are a wide variety of advising styles and different advisors will have different expectations for students. For example, some advisors may expect more student

independence, while others will want more interaction with their advisees. The type and level of interaction with your advisor will likely change as you matriculate through the program. For example, students at the beginning of their graduate program might meet with their advisors more sporadically; only as they have specific questions they need answered, while a more advanced student might be meeting weekly with an advisor as they propose and clarify their research.

Advising expectations

Successful graduate students communicate with their advisors and solicit relevant guidance throughout their entire graduate program. At a minimum, students should confer with their advisors about their coursework and at each milestone in their graduate program. This is critical, as advisors are often required to approve student decisions with their signature (e.g., degree plan, examination committees). They can often—but not always—provide guidance regarding the policies and procedures that are outlined in the various handbooks and policy documents for the track, program, college and university. (The staff in 250 EdSciB are often more helpful for this.) Advisors can also help students navigate things like attending or presenting at conferences/colloquia, putting together an academic CV, etc. Resources for faculty and students are available at the *Graduate and Professional Student Advising* website (http://gradvising.umn.edu/).

Registration

To maintain their active status, all graduate students must register every fall and spring semester. Those who do not register each semester are considered to have withdrawn and their Graduate School records are deactivated. Inactive students may not register for courses, take examinations, submit degree plan or thesis proposal forms, file for graduation, or otherwise participate in the University community as Graduate School students. Those who wish to resume graduate work must request readmission to the Graduate School and the program and, if readmitted, must register in the Graduate School for the term of readmission to regain their active status. Students who are readmitted will be responsible for the course and exam requirements in effect as of the readmission term.

Note: Students must register before the beginning of a term to avoid late fees. Faculty in QME are often not available during the summer term, and students should plan to complete their summer and fall term registrations before the end of the spring term.

Funding

Graduate students fund their education through a variety of ways. The most common form of support is through research and teaching assistantships; these positions provide a salary and may also provide a tuition reduction and health insurance. Your advisor may have suggestions of assistantships for which you might qualify. You should also inquire about positions in various research centers (e.g., Center for Applied Research and Educational Improvement; CAREI) and other departments (e.g., Institute of Child Development) that might hire research assistants. Also see the *Graduate Assistant Employment Services* webpage to search for assistantships (https://humanresources.umn.edu).

If you are interested in a teaching assistantship, talk to Andrew Zieffler, Elizabeth Fry, or Robert delMas. If you are interested in TAing for a specific class, you should talk with the instructor of that class. Your advisor may also be aware of courses that need a teaching assistant.

Fellowships, scholarships, and grants

There are many scholarships, grants, and fellowships available through the University and other organizations (such as professional and fraternal organizations). More information about University fellowships are available at the *Funding & Tuition* page at https://www.grad.umn.edu.

Note that some of these awards require nominations by the graduate program or department. When that is the case, the department/program will often set an earlier application deadline than is listed on the University website (so that the Director of Graduate Studies and the Graduate Advisory Committee may select the most competitive applications to forward for University-wide consideration). Check with your advisor or the DGS for more information on these internal deadlines.

URLs and resources for QME students

In addition to this handbook, all QME students should read the *University of Minnesota Graduate School Catalog* (http://www.catalogs.umn.edu/grad/) and the *Educational Psychology Graduate Program Handbook* (http://www.cehd.umn.edu/edpsych/current/qme/). These documents provide more specific information about degree requirements and university policies.

Below are links to several other useful resources:

- **QME—Student Resources:** Links to handbooks, departmental and program forms, etc. http://www.cehd.umn.edu/edpsych/current/qme/
- **Graduate Student Services and Progress (GSSP):** Online university portal to register for many of the program milestones (e.g., written and oral examinations) and assign/update committee members. https://onestop.umn.edu/academics/graduate-student-services-and-progress-gssp
- **Onestop Student Services Forms:** An A–Z index of university forms you may need. Also, browse by topic. https://onestop.umn.edu/forms
- **Graduate Student Catalogs:** General and program specific information, university policies, services, etc. http://www.catalogs.umn.edu/download/index.html
- **Onestop Academics:** Portal to information on everything academic related including academic planning, leaves, graduation, grades, transcripts, and more. https://onestop.umn.edu/academics

Studies for the master's (MA) degree

General information for MA students

The MA is awarded in recognition of academic accomplishment in QME as demonstrated by a successful completion of coursework, preparation of a Plan B paper, and passing of the final oral examination. The MA curriculum for QME students is available at http://www.cehd.umn.edu/edpsych/programs/qme/.

Students successfully completing the MA degree are capable of participating in research of an applied nature, developing tests and surveys, translating research findings for application to educational settings, and assisting professional educators with measurement, evaluation, and statistical issues.

Length of program

The MA program in Educational Psychology with an emphasis in QME requires successfully completing the QME and Educational Psychology core courses, and writing a Plan B research paper. With normal progress, most students should be able to complete the degree in four terms of full-time study.

Grade point average (GPA)

The Graduate School requires master's students to maintain a grade point average (GPA) of at least 2.80 on a 4.0 scale (A = 4.00, B = 3.00, C = 2.00, D = 1.00, and F = 0.0). Grades of C- or better and S (Satisfactory) are acceptable on the official degree plan, but grades of S are not calculated in the GPA. At least two-thirds of the course credits included on the student's *Degree Plan* must be taken A–F. All EPSY and QME core courses must be taken A–F as well.

Time limit for degree completion

The Graduate School requires all master's work be completed within a period of five years after initial enrollment in the program. For more information see: https://policy.umn.edu/education/mastersperformance.

Coursework

QME master's students take 33 credits distributed as follows:

- 12 credits EPSY core requirements;
- 18 credits QME core requirements (6 credits may be used to satisfy EPSY core requirements);
- Two additional courses (minimum 6 credits) in the area of emphasis are determined in consultation with advisor; and
- 3 Plan B paper credits (EPSY 5991).

All EPSY and QME core courses must be taken with the A–F grading option. Additional information about required QME coursework is available at: http://www.cehd.umn.edu/edpsych/programs/qme/

QME core requirements

All QME master's students are required to take the following core courses:

- EPSY 5221—*Principles of Educational and Psychological Measurement*
- EPSY 5243—*Principles and Methods of Evaluation*
- EPSY 5244—*Survey Design, Sampling, and Implementation*
- EPSY 5247—*Qualitative Methods in Educational Psychology*
- EPSY 8251—*Statistical Methods in Education I*
- EPSY 8252—*Statistical Methods in Education II*

Educational Psychology core requirements

QME master's students must satisfy the following Educational Psychology core requirements:

- 3 credits in learning/cognition;
- 3 credits in social psychology/personality;
- 3 credits in statistics;

- 3 credits in measurement; and
- 3 Plan B paper credits (EPSY 5991)

Note: The QME core courses EPSY 8251 or 8252 will satisfy the Educational Psychology core statistics requirement. And, the QME core courses EPSY 5221, 5243, 5244 or 5247 will satisfy the Educational Psychology core measurement requirement.

Transfer of credits and course waivers

Master's students may transfer up to 12 credits of graduate coursework taken either at another institution or at the University of Minnesota prior to admission may be used towards a master's program if approved by the program faculty and the Graduate School. Courses must be taken for graduate-level credit (after the award of the undergraduate degree), and appear on official graduate school transcripts. Courses that are officially transferred count toward both the coursework *and* credit requirements.

In rare circumstance coursework may also be waived (without transferring graduate-level credits) because of previous coursework, experience, or other proof of competence. In this case, the waived coursework *does not change the total number of credits a student must earn* for the master's degree.

Students must complete the *Department of Educational Psychology Internal Petition Form* to waive any coursework (transferred or otherwise). This form is available at: http://www.cehd.umn.edu/edpsych/forms/.

Degree plan

Master's students are required to file an official *Degree Plan* online after completing 10 credits (ordinarily no later than the second semester of the first year). To view and submit your MA program, access the *Grad Planning & Audit System* (GPAS) through MyU (https://www.myu.umn.edu). In MyU, select the "Academics" tab, and then select the "Grad Planning & Audit System" link at the bottom of the page. For instructions see https://onestop.umn.edu/academics/gpas.

Students should list all coursework, completed and proposed, that will be offered in fulfillment of degree requirements, including transfer work (see "Transfer of credits and course waivers" section). Revisions to a degree plan can be made with your advisor's approval by sending an email to the DGS Assistant.

Examination committee

In collaboration with their advisor, MA students must identify and recruit approved graduate faculty to participate on their examination committee. This should be done during the student's second semester of study.

The final examination committee consists of at least three members: at least two who represent QME and at least one who represents a field outside of QME. All committee members must have Graduate Education Responsibilities at the University of Minnesota. The following rules also apply to forming your examination committee:

- **Advisor:** Your advisor will be one of the members of your examination committee and must be a member representing QME.
- **Outside Member:** At least one committee member must represent a field outside of QME. This member may be from another program area in Educational Psychology. If the student has a declared minor(s), *the outside member(s) must be from the minor field(s).*
- **Co-advisor:** If you have a co-advisor, that person must be a member of your examination committee. Your co-advisor may be counted as a member that represents QME or as a member representing a field outside of QME.

Note: Members cannot satisfy the requirement with respect to more than one field.

When choosing your committee, you will want to select committee members who have knowledge and interests aligned with your research interests. It is important that you contact potential committee members to ask them to serve on your committee. All members of the committee and the student must participate in the final examination.

Assigning members to your committee

Once you have decided on a committee, you need to identify which committee members will serve in the following roles:

- Outside/minor field examiner
- Chair of the committee (Your advisor may serve in this role.)

You can then submit your examination committee and assign roles at https://onestop.umn.edu/academics/examination-committees. At this same website, you can also determine whether potential committee members have Graduate Education Responsibilities at the University of Minnesota and also update/change your examination committee.

Plan B paper

Students demonstrate their knowledge of research and scholarship by working independently on a research project and writing up this research into a Plan B paper. Typically, this paper resembles a thesis in length and format; however, it differs from a thesis in that *it need not represent an original, empirical study*. For example, a student may prepare a comprehensive review of research that reviews, organizes, synthesizes, and critiques previously conducted research and scholarly discussion pertaining to issues in the student's area of concentration. A paper that involves the analysis, or re-analysis, of previously collected data may also satisfy this requirement.

Proposal of Plan B paper topic

Once the student and advisor have identified a topic for the Plan B paper, the student will write a proposal for the topic to be reviewed by the reviewers on the examination committee. This proposal should be 1–3 pages in length, and describe the topic of the paper, research question(s), and proposed methodology.

Once the proposal has been approved, the student writes the Plan B paper under the advisor's supervision. A suggested format for the Plan B paper appears in Appendix A.

Note: Two faculty members from QME and one faculty member who represents a field outside of QME, who are on your examining committee serve as reviewers (readers).

Final oral examination

The purpose of the final oral exam is to judge the student's overall qualifications for being conferred the MA degree. While the nature and content of the final oral examination is determined by the examination committee members, it is typically, focused on the student's Plan B paper (and on the coursework listed on the student's *Degree Plan*). Students should also be prepared to talk about their career direction and what they plan to do once the MA is conferred.

The final oral examination is a closed examination, open only to the examination committee and the student. All examination committee members, and the student, must be present at the final oral examination. The absence of any member results in an invalid examination.

Note: Students must provide the reviewers with a copy of the Plan B paper at least 14 days before the scheduled date of the final oral examination.

Evaluation of the final oral examination

There are two potential outcomes for the final oral examination: pass, or fail. A majority vote of an examining committee is required to pass the examination. If the student fails the final examination, that student may retake the examination only if all committee members, or all committee members save one, approve this option. The second attempt to pass the master's final examination must use the same committee members as the first examination unless an emergency situation necessitates a substitution. If the committee does not approve a retake, or if the student fails the second attempt, the student will be terminated from the program.

The examination committee may ask the student to revise the Plan B paper as a condition for passing the examination. If revisions to the Plan B paper are required as a condition of passing, the advisor(s) must certify that the revisions have been completed before the degree is awarded.

Form to submit

If the oral examination is successfully passed, the Examination Committee will sign the *Final Examination Report* form (available in the *Graduation Packet*; see "Graduation Requirements" section). Bring the signed *Final Examination Report* form to the office staff in 250 EdSciB before the last working day of the intended month of graduation. They will make a scanned copy for you and electronically submit the form to Graduate Student Services and Progress (GSSP).

Applying for Graduation

Students should obtain a *Graduation Packet* early in the semester they wish to graduate. This can be obtained from https://onestop.umn.edu/ by selecting the "Academics" tab and then clicking the "Checklists for grad students" link. This packet includes instructions for filling out the online *Application for Degree* and also includes the *Final Examination Report* form. The *Application for*

Degree must be submitted online through MyU by the first working day of the month you intend to graduate. The *Final Examination Report* will need to be signed by your Examination Committee and submitted to the office staff in 250 EdSciB. More information is available at https://onestop.umn.edu/academics/apply-graduate

Commencement

The College of Education + Human Development hosts commencement exercises each year in May. It is not required but we do encourage all students who qualify to participate. To participate, MA students must have an approved Degree Plan on file. Typically the Department of Educational Psychology will also host a reception for graduating students and their families prior to commencement.

Note: RSVP-ing for the commencement ceremony has no bearing on your application to graduate. To actually apply to graduate, you need to submit the Application for Degree form to MyU.

Studies for the doctoral (PhD) degree

General Information for PhD students

The doctor of philosophy degree (PhD) is awarded chiefly in recognition of high attainment and ability in QME as demonstrated by a successful completion of coursework, passing of a written and oral qualifying examination, and preparation and defense of a dissertation based on original research that makes a significant contribution to knowledge in the student's field.. The PhD curriculum for QME students is available at http://www.cehd.umn.edu/edpsych/programs/qme/.

Students successfully completing the doctoral degree have an understanding of the content, methods, theories, and professional ethics associated with research methodology. They are capable of independently conducting research of an applied nature, translating sophisticated research findings for application in educational settings, and assisting educators with analyses and interpretations of research findings. Students also will have the skills to conduct methodological research, contributing directly to methodological practice and knowledge.

Length of program

The PhD program in Educational Psychology with an emphasis in QME requires successfully completing the QME and Educational Psychology core courses, writing a pre-dissertation paper, passing a written and oral preliminary (qualifying) exam, completing 24 thesis credits, and successfully proposing, writing, and defending a dissertation.

Students entering the doctoral program without a master's degree typically complete the program in 5–6 years of full-time study. Students entering the program with a master's degree in a related field typically complete the program in 3–4 years of additional full-time study. Students entering the program with a master's degree from the QME program can typically complete the program in 2–3 years of additional full-time study, since these students can count 30 credits from their MA program toward the PhD.

Grade point average (GPA)

The Graduate School requires doctoral students to maintain a grade point average (GPA) of at least 2.80 on a 4.0 scale (A = 4.00, B = 3.00, C = 2.00, D = 1.00, and F = 0.0). Grades of C- or better and S (Satisfactory) are acceptable on the official degree plan, but grades of S are not calculated in the GPA. At least two-thirds of the course credits included on the student's *Degree Plan* must be taken A–F. All EPSY and QME core courses must be taken A–F as well.

Note: Students pursuing a doctoral degree must register for doctoral thesis credits (EPSY 8888); these registrations are not graded and therefore cannot be used to meet the GPA requirements.

Time Limit for Degree Completion

The Graduate School requires all doctoral work be completed within a period of eight years after initial enrollment in the program.

Coursework

QME doctoral students take 72 credits distributed as follows:

- 27 credits of Educational Psychology core requirements;
- 9 credits of Educational Psychology electives;
- 18 credits of QME core requirements (6 credits may be used to satisfy Educational Psychology core requirements);
- 12 additional credits in the area of emphasis (determined in consultation with advisor); and
- 24 thesis credits (EPSY 8888).

All Educational Psychology and QME core courses must be taken with the A–F grading option.

Educational Psychology core requirements

QME doctoral students must satisfy the following Educational Psychology core requirements:

- 9 credits in at least two areas of learning/cognition, social psychology/personality or history/systems;
- 9 credits in research methods (EPSY 8215, 5244 and 5247)
- 24 thesis credits (EPSY 8888).

QME core requirements

All QME doctoral students need to take the following core courses:

- EPSY 5221—*Principles of Educational and Psychological Measurement*
- EPSY 5243—*Principles and Methods of Evaluation*
- EPSY 8251—*Statistical Methods in Education I*
- EPSY 8252—*Statistical Methods in Education II*
- EPSY 8264—*Advanced Multiple Regression Analysis*
- EPSY 8xxx—8xxx-level measurement course (minimum 3 credits)

Additional information about required QME courses is available at: http://www.cehd.umn.edu/edpsych/programs/qme/

Note: The QME core courses may be used to satisfy the 9 elective credits in the Educational Psychology core requirements.

Transfer of credits and course waivers

Doctoral students may transfer an unlimited number of credits of graduate coursework (taken at another institution or at the University of Minnesota prior to admission) towards their PhD program. However, these credits need to be approved by the QME faculty and the Graduate School. Courses must be taken for graduate-level credit (after the award of the undergraduate degree), and appear on official graduate school transcripts. Courses that are officially transferred count toward both the coursework *and* credit requirements.

In rare circumstance coursework may also be waived (without transferring graduate-level credits) because of previous coursework, experience, or other proof of competence. In this case, the waived coursework *does not change the total number of credits a student must earn* for the master's degree.

Students must complete the *Department of Educational Psychology Internal Petition Form* to waive any coursework (transferred or otherwise). This form is available at: http://www.cehd.umn.edu/edpsych/forms/.

Note: Doctoral students must take a minimum of 12 course credits at the University. Transferred credits can include a maximum of 12 graduate course credits taken as non-degree seeking or non-admitted status. Transfer of thesis credits is not allowed.

Degree plan

Within their first four semesters of study in the doctoral program, PhD students are required to file an official *Degree Plan* online. To view and submit your PhD program, access the *Grad Planning & Audit System* (GPAS) through MyU (https://www.myu.umn.edu). In MyU, select the "Academics" tab, and then select the "Grad Planning & Audit System" link at the bottom of the page. For instructions see https://onestop.umn.edu/academics/gpas.

Students should list all coursework, completed and proposed, that will be offered in fulfillment of degree requirements, including transfer work (see "Transfer of credits and course waivers" section). Revisions to the *Degree Plan* can be made with your advisor's approval by sending an email to the DGS Assistant.

Note: The Department of Educational Psychology requires doctoral students to submit their Degree Plan prior to taking the Written Preliminary Exam.

Examination committees: Preliminary oral and final oral

In collaboration with their advisor, PhD students must identify and recruit approved graduate faculty to participate on their *Preliminary Oral Exam Committee*. This should be done during the student's second or third year of study.

Students must also identify and recruit approved graduate faculty to participate on their *Final Oral Examination Committee*. The *Final Oral Examination Committee* is not required to include the same members who served on the *Preliminary Oral Exam Committee*. This committee should be identified after successfully completing the oral preliminary exam.

Both committees consist of at least four members: at least three who represent QME and at least one who represents a field outside of QME. All committee members must have Graduate Education

Responsibilities at the University of Minnesota. The following rules also apply to forming both of your examination committees:

- **Advisor:** Your advisor will be one of the members on your examination committees and must be a member representing QME.

- **Outside Member:** At least one committee member must represent a field outside of Educational Psychology. If the student has a declared minor(s), the outside member(s) *must be from the minor field(s)*.

- **Co-advisor:** If you have a co-advisor, that person must be a member of your examination committee.

Note: Members cannot satisfy the requirement with respect to more than one field. QME students may not use QME faculty members with joint graduate school appointments or external affiliations to fulfill the external membership requirement on graduate committees.

When choosing a committee, you will want to select committee members who have knowledge and interests aligned with your research interests. It is important that you contact potential committee members to ask them to serve on your committee. All members of the committee and the student must participate in the final examination.

Assigning members to your committee

Once you have decided on a committee, you need to identify which committee members will serve in the following roles:

- Outside/minor field examiner
- Chair of the committee (Your advisor or co-advisor may serve in this role for the prelim oral exam, but not for the final oral exam.)

You can then submit your examination committee and assign roles at https://onestop.umn.edu/academics/examination-committees. At this same website, you can also determine whether potential committee members have Graduate Education Responsibilities at the University of Minnesota and also update/change your examination committee.

Pre-dissertation research project

Students must successfully complete a pre-dissertation research project under the supervision of their adviser. This project, which entails a qualitative or quantitative analysis of empirical data, needs to include a relevant review of research literature, methodology, analytic results, and a brief discussion. The paper may be a synthesis of existing work, pilot study, and/or may be based on extant data.

If a student has completed a Master's thesis in which an empirical data analysis was conducted, this may satisfy the pre-dissertation research project requirement (check with your advisor). Students who complete an MA in QME will automatically satisfy this requirement.

The pre-dissertation research project is a collaborative effort between you and your adviser, but if the paper were to be published, the student should be listed as first author of the paper.

To stay on track for timely completion of the degree, students should complete this requirement within the first two to three years in the program. An example timeline for timely completion is:

- **Year 1 (2nd Semester):** Identify topic and begin work on the review of relevant research.
- **Year 2 (1st Semester):** Develop/plan the methodology
- **Year 2 (2nd Semester):** Conduct the study and write up the results during their fourth term of doctoral study.

The pre-dissertation research paper typically has a structure similar to the outline presented in Appendix A. This paper should be formatted using the guidelines from the most recent *APA Style Guide*, and, as a guideline, should not exceed 25 pages in length.

Evaluation of the pre-dissertation research project

The written paper you produce from your pre-dissertation research project is evaluated by the *Pre-Dissertation Committee*, which is comprised of the adviser and one other faculty member in QME with graduate status. The two readers judge the project to be either acceptable or not acceptable. When judging the project, the evaluators take into account both the quality of the research and the quality of the written paper.

The pre-dissertation research project must be judged as acceptable for the student to continue in the program. If the project is judged as unacceptable, the student may revise and resubmit the project at the discretion of the committee.

Form to submit

When the project has been evaluated and approved by the *Pre-Dissertation Committee*, the will sign off on the *Pre-dissertation Research Project Completion* form. This form is available at http://www.cehd.umn.edu/edpsych/current/qme/. The signed form then needs to be submitted to the program staff in 250 EdSciB.

Written Preliminary Exam

All Ph.D. students in QME need to complete the written preliminary exam. In QME, there are two parts to the written preliminary exam:

1. Sit-down specialty examination; and
2. Writing a potentially publishable research paper.

Prior to undertaking the written preliminary exam, students need to have successfully:

- Completed the following three research methods courses:
    - *Survey Design, Sampling, and Implementation* (EPSY 5244)
    - *Qualitative Research Methods* (EPSY 5247), and
    - *Advanced Research Methods* (EPSY 8215);
- Completed the pre-dissertation research project; and
- Filed a degree plan with the department.

Scheduling the written preliminary exam

Prior to completing either part of the written preliminary exam, students should register for the exam. Registration for the written preliminary exam is done through the student's advisor and the director of graduate studies. The *Prelim Registration Form* is available at http://www.cehd.umn.edu/EdPsych/Current/QME.html. Once the form has been signed by your advisor, it should be submitted to the DGS Assistant (Lori Boucher) in 250 EdSciB.

Part I. Sit-Down Specialty Exam

Note: This part of the written preliminary exam may be waived if the student has a GPA ≥ 3.5 in the core QME and the three required research methods courses completed.

The purpose of the sit-down specialty exam is to assess the student's breadth and depth of understanding in the core areas of quantitative methods in education: evaluation, measurement, and statistics. The exam consists of eight constructed-response items, four items in the student's concentration area and two items in each of the remaining two core areas. Students will select three of the four concentration items, and one item from each of the other areas to respond to. For example, the sit-down examination for a student concentrating in measurement would have the following test blueprint:

	Topic		
Number of items	Evaluation	Measurement	Statistics
On exam	2	4	2
Student responds to	1	3	1

The exam items will be written by a committee consisting of three QME faculty members. Students will have four hours to respond to the five items. Students may not bring any materials to the exam, and will type their responses into a word-processing program on a computer that will be provided.

Students taking the sit-down specialty exam are responsible for the material in all the QME core courses and the three required research methods courses. This includes, but is not limited to: course notes, textbooks, journal articles, and technical reports used in these classes. The exam will be tailored to the students taking the exam at a given time, given their area of interest.

Evaluation of the sit-down specialty exam responses

Student responses will be read and scored by the three QME faculty who wrote the exam questions. This committee will make a pass/fail decision. Students have two opportunities to pass the sit-down

specialty exam. If the student fails the exam twice that student will be terminated from the QME/Ed Psych program.

The evaluation of exam responses will be based on the following principles:

- Does the response constitute a logical argument exhibiting an awareness of the depth and breadth of issues related to the question?
- Does the response address a research base in responding to the question?
- Does the response demonstrate professionalism?

These principles will be considered as well as the accuracy and completeness of all responses.

Each of the three QME faculty scoring the exam will independently score items using a five-point scale (five being the highest possible score). The scores will then be averaged across items and raters for an overall total score. If the average total score is below 2.5, the student fails. If there are major discrepancies between the original raters, additional raters may be asked to score the exam. Students will typically be informed of their mark on the exam within one month of completing the exam.

If a scheduled exam is missed and not taken as scheduled, the student must petition the advisor or prelim committee for permission to take the sit-down specialty exam on a new date. Petitions will be reviewed on an individual basis. Typically, petitions are accepted only from those students facing extraordinary circumstances. If a petition is denied, the missed prelim will be recorded as a fail.

Part II. Writing a potentially publishable research paper

In the second part of the written preliminary exam, each QME student will write a potentially publishable research paper and submit it to the Doctoral Prelim Committee for evaluation. Although the paper does not actually have to be published for successful completion of this milestone, it needs to be of the same quality in terms of scholarship, writing, etc.

Note. A QME student who has already written and published, an appropriate research paper in the field of QME can petition their advisor to request that the already-published research paper be substituted for this requirement.

In conjunction with their advisor or another faculty member in QME, the student proposes a topic for the research paper. The topic of the paper should be agreed to by the student and their advisor before the student begins work on the paper.

Students may use their work from EPSY 8215 or their pre-dissertation research project as a basis for the research paper. However, the research paper cannot be limited to work in EPSY 8215 or the pre-dissertation research project. Rather, *the paper must be sufficiently different from other research papers the student has written to merit treatment as a new piece of work* in order to satisfy this part of the written preliminary exam.

In general, the structure of the paper will follow that outlined for the Master's Plan B research paper (see Appendix A). Data for the paper may be collected by the student as part of satisfying this requirement or may come from another source (e.g., data from a research project or an extant dataset in the public domain). There are no specific requirements for the length of the paper, but the paper should be formatted using style guidelines from the most current *APA Publication Manual*. (Students with an area of emphasis in Statistics Education should format their paper according to the *Guidelines for Authors* provided by the *Statistics Education Research Journal* (SERJ; https://iase-web.org/Publications.php?p=SERJ)

As a guideline, students should plan to complete the research paper within one semester. Once completed, students are strongly encouraged to submit their work for possible publication in an appropriate journal.

Evaluation of the potentially publishable research paper

Prior to submission for evaluation the *student's advisor is allowed to review the paper one time*. After the paper is submitted for evaluation, it will be read and evaluated by at least two QME faculty, one of whom will be the student's advisor. The readers will make a pass/fail decision. In the event of a split decision, a third member of the QME faculty will be appointed by the QME Program Coordinator to read and evaluate the paper. Students will typically be informed of their mark on the Part II paper within one month of submitting it for evaluation.

Students who fail this part of the written preliminary exam have one additional opportunity to resubmit their paper. This may entail substantial revision of the original paper or may require the writing of a new paper under the supervision of the advisor.

Forms to submit

When both parts of the written preliminary examination have been completed, and the written paper has been evaluated and approved, the written prelim committee will sign off on the *QME Written Specialty Exam Completion Form*. You also need to complete the *Checklist for Showing Completion of Written Prelim* and also have it signed by the written prelim committee. Both forms are available at: http://www.cehd.umn.edu/edpsych/current/qme/

Both signed forms then need to be submitted to the program staff in 250 EdSciB. The *Written Specialty Exam Completion Form* should be submitted to the QME Assistant (Sharon Sawyer), and the *Checklist for Showing Completion of Written Prelim* should be submitted to the DGS Assistant (Lori Boucher).

Oral preliminary exam

After successfully completing the written preliminary exam, students must pass an oral preliminary exam. The oral preliminary exam is used to assess a student's preparation to conduct dissertation work. In Educational Psychology, there are two components to the oral preliminary exam: (1) writing the orals paper, and (2) the oral examination.

For additional information on the oral preliminary exam, see the *Graduate Education General Information and Policies* catalog (http://www.catalogs.umn.edu/download/index.html) and the *Educational Psychology Graduate Program Handbook* (http://www.cehd.umn.edu/edpsych/current/qme/).

Orals paper

As part of the preparation for the oral examination, the student writes a scholarly document (orals paper) that synthesizes an area of theory, research, and/or practice closely related to the student's own academic program and area of focus. While not required, this topic is often closely related to the student's proposed dissertation topic. In fact, for many students, the orals paper will form the basis for the *Review of the Literature* chapter in their dissertation.

The Educational Psychology Handbook (p. 10) describes orals papers in this way:

> The topic and range of coverage of the Prelim Oral Paper is selected by the student with approval of his/her advisor. The intent of the paper is to integrate the literature in a prescribed area of research. It should not include a specific plan of the student's dissertation.
>
> The paper is modeled on those integrative reviews found in the *Review of Educational Research*, *Psychological Bulletin*, or *Psychological Review*. Conceptually, the paper is neither a mindless listing of studies in excessive detail, nor a widely speculative paper with no evidence offered in support of argument. It is not flawed by theoretical gaps in the literature review. Rather, the paper identifies major issues in the area chosen. It is guided by ideas and uses information selectively and critically with respect to those ideas. It reflects methodological sophistication in the description of research strategies and results. The paper is to be coherent. Being analytic, integrative, and topically constrained, the paper is not to be discursive or of excessive length. It will normally range from 20-40 pages of double-spaced text.

Although the topic for the orals paper will be developed by the student in consultation with the advisor, the student must assume primary responsibility for selection of a topic, as well as, the generation of scope and procedural plans for conducting the review, collection of relevant literature and research information, review and synthesis, and preparation of the final paper. The student may receive collegial support and feedback from the advisor or others ancillary to this project, but the final paper must reflect the individual effort and perspectives of the student. In short, the orals paper is a single-author effort.

Oral examination

The oral examination is completed by all University of Minnesota doctoral students. This examination is taken "after completing a substantial portion of the coursework and passing the preliminary written examination[s]" (Graduate School Catalog, 2009, p. 24). The examination "covers the major field, the minor field or supporting program, and any work fundamental to these areas, including possible plans for dissertation research" (Graduate School Catalog, p. 25). This examination is conducted, in a closed session, by a four-member *Oral Prelim Examining Committee*.

Scheduling the oral examination

With the advisor's approval, typically once the orals paper is nearly complete, the student can schedule a date for their oral examination. This is scheduled as a two-hour block of time, and may take some time to coordinate the Examining Committees' schedules. You must also formally schedule the preliminary and final oral examination with Graduate Student Services and Progress (GSSP) online as soon as the dates are set, but no later than one week prior to the examinations. You can do this at: https://onestop.umn.edu/academics/graduate-student-services-and-progress-gssp. At least two weeks prior to the scheduled exam date, the orals paper should be provided to the student's *Examining Committee* for review.

Note: "Preliminary oral examinations should not be scheduled during the summer unless the members of the assigned committee can be assembled without substitution" (Graduate School Catalog, 2009, p. 25).

Evaluation of the oral examination

While no specific criteria are provided for faculty committee's evaluations of student performance, prior experience suggests a set of skills and competencies that are important (but not necessarily sufficient) for successfully completing the oral examination. These skills represent accumulated training, experience, and perspective, and are fundamental not only to the oral examination but to other scholarly and professional endeavors.

- **Provide a clear statement of themes, findings, or dilemmas that represent your interest in an existing body of literature**
  - Make a clear statement of the area of literature, and theme(s) reviewed
  - Provide a rationale for, and support of, identified theme(s)
  - Describe the relation between identified theme(s), broader literature, and ongoing area of scholarly and/or professional activity
- **Have a sufficient store of facts**
  - Master foundational areas of content, theory, and technical skill
  - Read and discuss issues broadly
  - Maintain currency – monitor relevant journals and other periodicals, conference proceedings, and professional activities

- **Demonstrate your capacity** *to ask* **complex conceptual questions**
  - Identify gaps of knowledge or theory in extant literature
  - Identify gaps or shortcomings in methodology of extant literature
  - Integrate perspectives or findings from multiple areas or sources
- **Demonstrate your capacity** *to answer* **complex conceptual questions**
  - Define essential element(s) of questions, when asked
  - Identify information pertinent to each identified element
  - Organize and presenting information coherently
  - Check for understanding
- **Defend your own perspective, or change and accommodate that perspective when appropriate**
  - Develop clear, supportable perspectives and opinions on body of theory, research, and methods
  - Know what you think, and the assumptions and facts on which your perspectives and opinions are based
  - Identify views that diverge from your own
    - When divergence can be accounted for by differences in factual knowledge, change your view to accommodate new facts
    - When divergence can be accounted for by differences in perspective or opinions, identify essential differences and provide rationale for your position on these differences
  - Be consistent, coherent, and respectful in responses to others' questions and challenges

There are three potential outcomes for the oral examination: pass, pass with reservations, or fail. Passing the preliminary oral exam allows the student to begin work on their dissertation and register for thesis credits. If the student passes the examination with reservations, the student is informed immediately, but the committee is permitted one week in which to convey its reservations to the student in writing, informing the student of the steps that must be taken to remove them. Once the student has satisfied the committee's reservations, the committee will inform the Graduate School which will inform the student that the reservations have been removed and that the student may proceed toward the doctoral degree.

Students who fail the oral exam have one additional opportunity to retake the exam. Attempting to pass the preliminary oral exam a second time may entail substantial revision of the original

preliminary oral paper or may require the writing of a new paper. Students who fail the oral examination a second time will be excluded from candidacy for the doctoral degree.

Note: At least 10 weeks need to pass between the initial oral examination and the retake.

Form to submit

If the oral examination is successfully passed, the *Examination Committee* will sign the *Preliminary Oral Examination Report* form. Bring the signed *Preliminary Oral Examination Report* form to the office staff in 250 EdSciB no later than the first workday after the examination. They will make a scanned copy for you and electronically submit the form to Graduate Student Services and Progress (GSSP).

Doctoral dissertation fellowship

The Graduate School awards doctoral dissertation fellowships on a competitive basis to support doctoral students during the dissertation writing stage of their graduate career. Students must be nominated by their program. Competition for these awards is fierce but we do encourage all eligible students to apply. Educational Psychology students have been successful in receiving this fellowship. To be eligible for the award, students must have completed all coursework on their degree plans, passed the written and oral prelim exams, and have submitted the *Thesis Planning Panel* form.

Early in the spring semester, students will be asked to submit proposals to the program. The Graduate Advisory Committee will review the proposals and submit the most promising to the Graduate School.

Information on the doctoral dissertation and other fellowships is available at http://www.grad.umn.edu/fellowships.index.html.

Doctoral dissertation

Each student writes a dissertation that presents the results of the student's dissertation research. An appropriate dissertation research project involves significant, original, and independent research work that is grounded in a body of literature. It presents hypotheses tested by data and analyses and provides a contribution or advancement in the fields of quantitative methods in education. It is the responsibility of the student's *Final Oral Examination Committee* to evaluate the dissertation in these

terms and to recommend awarding the doctorate only if the dissertation is judged to demonstrate these qualities.

Prospectus

Once the student has identified a research topic for the dissertation, and the advisor has approved that topic, the student will write a prospectus paper that includes a rationale based on the substantive literature for the research, the research question(s), and proposed methodology. This prospectus paper should be 10–20 pages in length.

This document is then presented to the student's *Thesis Planning Panel*, which include the student's advisor and two other QME faculty members (typically members of the student's doctoral committee). If the panel agrees that that the proposed topic is appropriate for an Educational Psychology dissertation and that the proposed design and procedures show enough promise such that the student could pursue the investigation further, they will sign the *Thesis Planning Panel* form. This form is available at http://www.cehd.umn.edu/edpsych/forms/. Once signed, this form should be submitted to the DGS Assistant (Lori Boucher) in 250 EdSciB. *This form must be submitted the semester following the preliminary oral exam.*

Carrying out the research and writing the dissertation

After the *Thesis Planning Panel* has signed off on the prospectus, the student carries out the proposed research under the advisor's supervision and prepares the written dissertation. A suggested format for the dissertation appears in Appendix A.

Once the dissertation has been written and your academic advisor believes it to be ready, the dissertation is then submitted to the dissertation reviewers (readers) on your *Final Oral Examination Committee*. (At least two faculty members from QME and one faculty member who represents a field outside of Educational Psychology, who are on your *Final Oral Examination Committee* serve as reviewers.) If they agree that the dissertation is ready to defend, they will sign off on the *Reviewers Report* form (available in the *Graduation Packet*; see Graduation section).

Note: Students must provide reviewers with a copy of the dissertation at least 30 days before the scheduled date of the doctoral final oral examination. Every designated reviewer on the doctoral

dissertation reviewer's report must certify that the dissertation is ready for defense before the doctoral final oral examination may take place.

Form to submit

Bring the signed *Reviewer's Report* form to the office staff in 250 EdSciB. They will make a scanned copy for you and electronically submit the form to Graduate Student Services and Progress (GSSP). The signed *Reviewer's Report* form can be submitted to Graduate Student Services and Progress (GSSP) up to the day of the exam.

Once Graduate Student Services and Progress (GSSP) has received your signed *Reviewer's Report,* they will issue the *Final Examination Report* form which is used at your dissertation defense. If there is sufficient time, GSSP will send the *Final Examination Report* form to the chair of your *Final Oral Examination Committee* via campus mail. If, however, the reviewer's form is emailed to GSSP and there is not enough time to send the *Final Examination Report* via campus mail, GSSP will send the student an email telling them they can pick up the form from GSSP at 333 Bruininks Hall with a photo ID. If the student brings the reviewer's form to GSSP immediately before the exam, they will issue the final exam form to the student right then and there.

Final oral examination

All doctoral candidates are required to successfully defend their dissertation research in an oral examination. The final oral examination includes (1) a public presentation of the candidate's dissertation to the *Final Oral Examination Committee* and the invited scholarly community; and (2) a closed session (open only to the *Final Oral Examination Committee* and the candidate) immediately following the public presentation. The nature and content of the final oral examination is limited to the candidate's dissertation topic and relevant areas, although the candidate should also be prepared to talk about their career direction and what they plan to do once the PhD is conferred. Typically, this examination takes about two hours.

Note: All Final Oral Examination Committee members and the candidate must be present at the final oral examination. The absence of any member results in an invalid examination.

Evaluation of the final oral examination

There are two potential outcomes for the final oral examination: pass, or fail. Immediately after the closed session of the final oral examination, the candidate will be excused from the room. The *Final Oral Exam Committee* will then take a written, secret ballot. They will then have a discussion and a second, final vote is taken.

To be recommended for the award of the doctoral degree, candidates must receive a vote of "pass" with no more than one dissenting member of the *Final Oral Examination Committee*.

According to the rules defined by the graduate faculty, the student either passes or fails the *Final Oral Examination*. Retakes are not permitted, and (unlike the *Preliminary Oral Examination*) there is no provision for a "pass with reservations." The faculty examining committee has both the authority and the responsibility to fail a student whose dissertation or performance in the oral defense does not meet the standards for award of the doctoral degree.

Form to submit

If the oral examination is successfully passed, the *Final Oral Examination Committee* will sign the *Final Examination Report* form. Bring the signed *Final Examination Report* form to the office staff in 250 EdSciB no later than the first workday after the examination. They will make a scanned copy for you and electronically submit the form to Graduate Student Services and Progress (GSSP).

The *Final Oral Examination Committee* may ask the candidate to revise the dissertation as a condition for passing the examination. If this happens, the candidate will be notified in writing (most likely via email) of all required revisions to the doctoral dissertation as well as specifying a time limit for the submission of the revised doctoral dissertation that is within one week of the final oral examination. The student's adviser is responsible for ensuring that the student includes the appropriate modifications and required revisions in the final dissertation. In this case, the advisor may withhold their signature from the Final Examination Report form until all revisions have been made.

If the revisions are substantial and the Final Examination Report form will not be submitted within a week of the Final Oral Examination, the chair of your Final Oral Examination Committee will send an email to Graduate Student Services and Progress (GSSP) stating that the form is being held pending revisions.

All dissertation revisions need to be completed and approved by your advisor before the degree will be awarded.

Submitting the final copy of the dissertation

Your final copy of the dissertation must be formatted according to the University of Minnesota formatting and submission guidelines (see more on at the "Thesis/dissertation submission and formatting" link under the "Academics" tab at https://onestop.umn.edu/).

You are required to file a digital copy of the dissertation with the University in accordance with University standards. (There are several submission criteria at the "Thesis/dissertation submission and formatting" page at Onestop.) In addition to submitting your thesis/dissertation online, you must submit the following documents to Graduate Student Services and Progress (GSSP):

- *Thesis/Dissertation Signature Page*
- *University Digital Conservancy Deposit Agreement*

All required materials must be submitted to Graduate Student Services and Progress (GSSP) by the last working day of the intended month of graduation.

Note: Students may choose whether or not to request an embargo of the publication of the dissertation for a limited period of time. To do this fill out the Dissertation or Thesis Hold Request Form at https://onestop.umn.edu/forms.

Applying for Graduation

Students should obtain a *Graduation Packet* early in the semester they wish to graduate. This can be obtained from https://onestop.umn.edu/ by selecting the "Academics" tab and then clicking the "Checklists for grad students" link. This packet includes instructions for filling out the online *Application for Degree*, the *Reviewers Report* form, the *Thesis/Dissertation Signature Page*, and the *University Digital Conservancy Deposit Agreement*.

The *Application for Degree* must be submitted online through MyU by the first working day of the month you intend to graduate. The *Final Examination Report* will need to be signed by the reviewers (readers) on your *Final Oral Examination Committee* and submitted to the office staff in 250 EdSciB.

The *Thesis/Dissertation Signature Page* is signed by your advisor and submitted to GSSP after you make any final revisions required by your *Final Oral Examination Committee*. The final copy of your dissertation is submitted to ProQuest. The *University Digital Conservancy Deposit Agreement* is also submitted to GSSP.

Commencement

The College of Education + Human Development hosts commencement exercises each year in May. It is not required but we do encourage all students who qualify to participate. To participate, PhD students must have submitted the *Thesis Planning Panel* form and have advisor. Typically the Department of Educational Psychology will also host a reception for graduating students and their families prior to commencement.

Note: RSVP-ing for the commencement ceremony has no bearing on your application to graduate. To actually apply to graduate, you need to submit the Application for Degree at MyU.

Appendix A

Below is a sample format for the master's degree Plan B research paper, pre-dissertation research project, written prelim paper, and doctoral dissertation. The use of chapters is specific to the doctoral dissertation; in the master's degree Plan B research paper, pre-dissertation research project, and written prelim paper, the chapter titles would be section headers rather than distinct chapters. This is only provided as a sample format, other outlines may also be appropriate. Check with your advisor for more specific advice about how to organize your particular paper.

Title page

Title of the work, names of the author, degree(s) of the author, department, and university to which it is submitted, date it is submitted.

Acknowledgments

Acknowledge all academic and technical assistance you obtained at each stage of the study.

Abstract

500 word abstract summarizing the study and its findings.

Table of contents

List all preliminary sections, all chapter titles, all major headings and subheadings of each chapter, reference list, and appendices. Include list of tables and list of figures. Page number of the beginning of each section (subsection) listed is put in the right hand margin.

Chapter I. Introduction

1. General background introduction
2. Purpose of the study, rationale, theory
3. Statement of the problem (i.e., hypothesis to be tested or research questions to be investigated)
4. Limitations of the study
5. Definitions of special terms you will use

Chapter II. Review of the literature

1. Organize and synthesize previously conducted research and scholarly discussion related to the problem being investigated. The goal of this section is to show how each of your hypotheses or research questions are derived from the previous work of others.
2. End chapter with a summary section

Chapter III. Method

Write this chapter before conducting study and present it along with Chapters I and II as the research proposal. After the data analyses are complete, revise this chapter.

1. Sample
2. Procedures and research designs
3. Instruments used/developed and their reliability/validity
4. How each research question/hypothesis will be tested
5. State hypotheses
6. What analysis research design will be carried out?
7. How will the results be organized and presented?

Chapter IV. Results and analysis

1. General data analysis and results
2. Data results specific to each hypothesis are presented
3. Chapter review

Chapter V. Discussion, interpretation, conclusion, summary

Summarize the entire project including what hypothesis/questions were investigated, why they were investigated, how they were investigated, the major findings, and your conclusions.

1. Discuss the findings and the hypothesis in a holistic and integrated fashion.
2. Explain any extraneous factors that may have led to the results you obtained.
3. Discuss the practical and theoretical implications of your findings and precisely how your research supports each implication.

4. State the conclusions to be drawn from your entire study (including review of the literature and empirical findings; i.e., integrate everything).
5. Discuss suggestion for future research, next stages of research, what others might do to follow up on your study.

References

List every reference cited in the body of the paper in alphabetical order by last name of author. Use APA style for each entry. List only those references actually cited in the body of the paper.

Appendices

Include long data tables, copy of instrument(s) used, special computer program written especially for this study, supplementary illustrated material, letters soliciting subjects, consent forms, and other material that supplement the text but which are not appropriate for inclusion in the text.